

3.2.8	Haloalkanes
	Synthesis of chloroalkanes understand the reaction mechanism of methane with chlorine as a free-radical substitution reaction in terms of initiation, propagation and termination steps know that chloroalkanes and chlorofluoroalkanes can be used as solvents understand that ozone, formed naturally in the upper atmosphere is beneficial be able to use equations such as the following to explain why chlorine atoms catalyse the decomposition of ozone and contribute to the formation of a hole in the ozone layer $\text{Cl}^\bullet + \text{O}_3 \rightarrow \text{ClO}^\bullet + \text{O}_2 \quad \text{and} \quad \text{ClO}^\bullet + \text{O}_3 \rightarrow 2\text{O}_2 + \text{Cl}^\bullet$ know that chlorine atoms are formed in the upper atmosphere when energy from ultra-violet radiation causes C–Cl bonds in chlorofluorocarbons (CFCs) to break appreciate that legislation to ban the use of CFCs was supported by chemists and that they have now developed alternative chlorine-free compounds

Nucleophilic substitution	understand that haloalkanes contain polar bonds
	understand that haloalkanes are susceptible to nucleophilic attack, limited to OH^- , CN^- and NH_3
	understand the mechanism of nucleophilic substitution in primary haloalkanes
	understand that the carbon–halogen bond enthalpy influences the rate of hydrolysis
	appreciate the usefulness of these reactions in organic synthesis
Elimination	understand concurrent substitution and elimination (including mechanisms) in the reaction of a haloalkane (e.g. 2- bromopropane with potassium hydroxide) and the role of the reagent as both nucleophile and base
	appreciate the usefulness of this reaction in organic synthesis